29/08/2026

CODE-C
Phase-3

Aakash
Medical | IIT-JEE | Foundations

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MERIDIAN TEST FOR NEET-2027 (XII PASSED)

MM: 720 **Test-01C** **Time: 180 Mins.**

Answers

1. (1)	37. (3)	73. (4)	109. (3)	145. (2)
2. (1)	38. (4)	74. (4)	110. (4)	146. (1)
3. (3)	39. (1)	75. (3)	111. (1)	147. (2)
4. (2)	40. (3)	76. (4)	112. (4)	148. (4)
5. (3)	41. (2)	77. (3)	113. (3)	149. (2)
6. (3)	42. (3)	78. (1)	114. (3)	150. (1)
7. (3)	43. (3)	79. (4)	115. (1)	151. (2)
8. (4)	44. (4)	80. (4)	116. (4)	152. (4)
9. (3)	45. (3)	81. (2)	117. (2)	153. (1)
10. (3)	46. (2)	82. (2)	118. (4)	154. (1)
11. (4)	47. (2)	83. (3)	119. (2)	155. (2)
12. (1)	48. (3)	84. (2)	120. (3)	156. (4)
13. (3)	49. (1)	85. (3)	121. (4)	157. (4)
14. (4)	50. (2)	86. (3)	122. (4)	158. (3)
15. (4)	51. (4)	87. (3)	123. (4)	159. (1)
16. (4)	52. (3)	88. (2)	124. (3)	160. (2)
17. (1)	53. (1)	89. (3)	125. (4)	161. (3)
18. (1)	54. (1)	90. (2)	126. (1)	162. (4)
19. (3)	55. (1)	91. (4)	127. (2)	163. (3)
20. (4)	56. (2)	92. (2)	128. (4)	164. (4)
21. (2)	57. (4)	93. (2)	129. (1)	165. (2)
22. (2)	58. (2)	94. (3)	130. (3)	166. (1)
23. (3)	59. (1)	95. (2)	131. (4)	167. (2)
24. (1)	60. (3)	96. (2)	132. (3)	168. (2)
25. (4)	61. (2)	97. (3)	133. (3)	169. (1)
26. (4)	62. (4)	98. (1)	134. (3)	170. (3)
27. (3)	63. (4)	99. (3)	135. (3)	171. (2)
28. (4)	64. (2)	100. (4)	136. (2)	172. (3)
29. (2)	65. (2)	101. (4)	137. (1)	173. (3)
30. (4)	66. (1)	102. (4)	138. (3)	174. (1)
31. (4)	67. (3)	103. (4)	139. (3)	175. (3)
32. (2)	68. (1)	104. (2)	140. (4)	176. (4)
33. (1)	69. (1)	105. (3)	141. (1)	177. (1)
34. (3)	70. (2)	106. (3)	142. (3)	178. (4)
35. (4)	71. (3)	107. (3)	143. (4)	179. (2)
36. (3)	72. (3)	108. (2)	144. (4)	180. (4)

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Hints and Solutions

PHYSICS

(1) **Answer :** (1)
Solution:
$v = \frac{A\sqrt{x}}{x+B}$
[*B*] = [L]
$\left[\frac{AL^{\frac{1}{2}}}{L}\right] = \left[LT^{-1}\right]$
$\left[A\right] = \left[L^{\frac{3}{2}}T^{-1}\right]$
$\left[AB\right] = \left[M^0L^{\frac{5}{2}}T^{-1}\right]$

(2) **Answer :** (1)
Solution:
$s = ut + at^2$
[s] = [*L*]
[*ut*] = [*L*]
[at^2] = [*L*]
Hence, equation is dimensionally correct.

(3) **Answer :** (3)
Solution:
L.C. = 1 MSD – 1 VSD
1 MSD = 2 mm
Now 20 VSD = 16 MSD
1 VSD = $\frac{16}{20}$ MSD
∴ L.C. = 1 MSD – $\frac{8}{10}$ MSD
= $\left[1-\frac{8}{10}\right]$MSD $= \frac{2}{10}\times 2$ mm $= \frac{4}{10}$mm
= 0.4 mm

(4) **Answer :** (2)
Solution:
[Linear momentum] = [$M^1 L^1 T^{-1}$]
[Force] = [$M L T^{-2}$]
[Power] = [$M L^2 T^{-3}$]
[Density] = [$M L^{-3} T^0$]

(5) **Answer :** (3)
Solution:
Gravitation potential = $\frac{\text{Grav. Potential energy}}{\text{mass}}$
$[V] = \left[\frac{\text{Energy}}{\text{mass}}\right] = \left[\frac{FL}{M}\right] = \left[\frac{\left(\frac{Mv}{t}\right)L}{M}\right] = \left[v\times v\right] = \left[v^2\right]$

(6) **Answer :** (3)
Hint:
KE $= \frac{1}{2}mv^2$
Solution:
Unit of KE = Mass $\left[\frac{\text{Length}}{\text{Time}}\right]^2$

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$= 100\ \text{g} \times \left(\frac{1\ \text{cm}}{10\ \text{s}}\right)^2$

= 1 g cm^2/s^2 = 1 erg

(7) **Answer :** (3)
Hint:
$\frac{\Delta s}{s} = 2\frac{\Delta r}{r}$
Solution:
$\frac{\Delta s}{s} \times 100 = 2 \times \frac{\Delta r}{r} \times 100$
= 2 × $\frac{0.2}{20.0} \times 100$
= 2%

(8) **Answer :** (4)
Solution:
Pure numbers like 1, 2, 3... and arguments of special functions like trigonometric, are not deduced by dimensional method.
The addition of two term cannot be deduced by dimensional method.

(9) **Answer :** (3)
Solution:
Random error $\propto \frac{1}{\text{Number of observation}}$

(10) **Answer :** (3)
Solution:
Stress = $\frac{\text{Force}}{\text{Area}}$
has dimensions $\frac{[\text{MLT}^{-2}]}{[\text{L}^2]}$
$= \left[\text{ML}^{-1}\,\text{T}^{-2}\right]$

(11) **Answer :** (4)
Hint:
$v = \frac{dx}{dt}$ and $a = \frac{d^2x}{dt^2}$
Solution:
$x = 5t^2 + 4t + 8$
$\Rightarrow x(t = 2\ \text{s}) = 5 \times 2^2 + 4 \times 2 + 8$
= 36 m
Now, $v = \frac{dx}{dt}$
$\Rightarrow v = 10t + 4$
$v\ (t = 0) = 4$ m/s
$v(t = 3\ \text{s}) = 10 \times 3 + 4 = 34$ m/s
Now, $a = \frac{dv}{dt} = 10\ \text{m/s}^2$

(12) **Answer :** (1)
Solution:
$v_{avg} = \frac{3x}{\frac{x}{v_1} + \frac{x}{v_2} + \frac{x}{v_3}}$
$= \frac{3v_1v_2v_3}{v_2v_3 + v_1v_3 + v_1v_2}$

(13) **Answer :** (3)
Solution:
$a = 2t^{3/2}$
$\frac{dv}{dt} = 2t^{3/2}$
$\int_u^v dv = 2\int_0^t t^{3/2}dt$
$v - u = 2\left|\frac{t^{5/2}}{\left(\frac{5}{2}\right)}\right|_0^t$
$v = u + \frac{4}{5}t^{5/2}$

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(14) Answer : (4)
Solution:
$P = a + bt^2$
$\frac{dp}{dt} = 2bt$
$F = 2bt \Rightarrow F \propto t$

(15) Answer : (4)
Solution:
$\frac{dx}{dt} = 2\sqrt{x}$
$\int_0^4 \frac{dx}{2\sqrt{x}} = \int_0^t dt$
$\left[\frac{x^{-\frac{1}{2}+1}}{1}\right]_0^4 = \int_0^t dt$
$t = \left(\sqrt{4} - 0\right)$ = 2 s
$\therefore v_{av} = \frac{x}{t} = \frac{4}{2}$ = 2 m/s

(16) Answer : (4)
Solution:
$\frac{dy}{dx}$ is zero at *A* and *C*

(17) Answer : (1)
Solution:
$v = x^{3/2}$
$a = \frac{vdv}{dx} = x^{3/2} \cdot \frac{3}{2} \cdot x^{1/2} = \frac{3}{2}x^2$
$a \propto x^2$

(18) Answer : (1)
Solution:
$h = \frac{u^2}{2g} = \frac{(20)^2}{2\times 10} = 20 \text{ m}$
$t = \frac{u}{g}$
$\Rightarrow t = \frac{20}{10} = 2 \text{ s}$
$\Rightarrow h = 20 \text{ m}, \quad t = 2 \text{ s}$

(19) Answer : (3)
Solution:
$\frac{h}{4} = \frac{1}{2}g \cdot \left(t-1\right)^2$
$h = \frac{1}{2}gt^2$
$\frac{\frac{1}{2}gt^2}{4} = \frac{1}{2}g(t-1)^2$
$t^2 = 4(t-1)^2$
$t = 2(t-1)$
$t = 2$ s

(20) Answer : (4)
Solution:
For constant acceleration, velocity of the particle is directly proportional to time with constant slope.
For 0 to T/2 interval, slope of v-t graph is more than interval T/2 to T.

(21) Answer : (2)
Solution:
$S_1 = \frac{1}{2}at^2 = \frac{1}{2} \times 2 \times 10^2 = 100 \text{ m}$
$v = 2 \times 10 = 20$ m/s
$\therefore \quad S_2 = 200 \text{ m}$
$v_{av} = \frac{300}{20}$ m/s

(22) Answer : (2)

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Solution:
Comparing the given equation with the standard equation of trajectory of projectile

$y = x\tan\theta - \frac{x^2\tan\theta}{R}$

$H = \frac{R\tan\theta}{4} = \frac{16\times 80}{4}$

= 320 m

(23) **Answer :** (3)
Solution:
A unit vector has magnitude equal to unity as well as direction associated with it.

(24) **Answer :** (1)
Solution:

$\frac{a_1}{a_2} = \frac{r_2}{r_1} = \frac{12}{4} = 3$

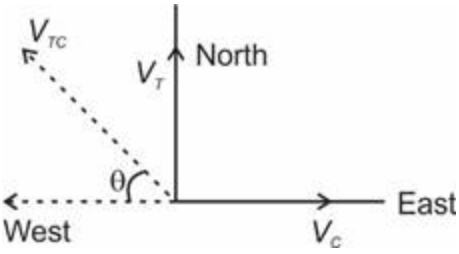
(25) **Answer :** (4)
Solution:
For uniform circular motion, tangential acceleration should be zero whereas centripetal acceleration is non-zero.

(26) **Answer :** (4)
Solution:

$H = \frac{u^2\sin^2\theta}{2g} \Rightarrow H_B > H_A$

$T = \frac{2u\sin\theta}{g} \Rightarrow T_B > T_A$

(27) **Answer :** (3)
Solution:



$\vec{V}_{TC} = \vec{V}_T - \vec{V}_C$

$|V_{TC}| = \sqrt{V_T^2 + V_C^2}$

$= \sqrt{6400 + 3600}$

V_{TC} = 100 km h^{-1}

(28) **Answer :** (4)
Hint:
Use $\vec{S} = \vec{u}t + \frac{1}{2}\vec{a}t^2$
Solution:
Since initial position vector is zero, therefore

$\vec{r} = \vec{u}t + \frac{1}{2}\vec{a}t^2 \Rightarrow \vec{r} = 0 + \frac{1}{2}\times\left(7\hat{i} + 3\hat{j}\right)\times 4$

$\vec{r} = \left(14\hat{i} + 6\hat{j}\right)$m

(29) **Answer :** (2)
Solution:

$\left|\vec{A} + \vec{B}\right| = \left|\vec{A} - \vec{B}\right|$

If θ is angle between them,

$\therefore A^2 + B^2 + 2AB\cos\theta = A^2 + B^2 - 2AB\cos\theta$

$4AB\cos\theta = 0$

$\cos\theta = 0$

$\theta = \frac{\pi}{2}$

(30) **Answer :** (4)

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Solution:

Given $y = x - \frac{x^2}{80}$

On comparing with $y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$

$\tan\theta = 1 \Rightarrow \theta = 45^\circ$

Now, $\frac{g}{2u^2\cos^2\theta} = \frac{1}{80}$

$\Rightarrow \frac{10}{2u^2\times\left(\frac{1}{2}\right)} = \frac{1}{80}$

$u = \sqrt{800} = 20\sqrt{2}$ m s^{-1}

• Maximum height gained

$H_{\max} = \frac{u^2\sin^2\theta}{2g} = 20$ m

• Horizontal range

$R = \frac{u^2\sin 2\theta}{g} = 80$ m

(31) Answer : (4)

Solution:

$v^2 = u^2 + 2as$ (in y direction)

$4 = u^2 - 2 \times 10 \times 0.6$

$u^2 = 16$

$u = 4$

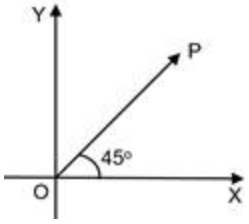
$\overrightarrow{v_i} = \left(6\hat{j} + 4\hat{j}\right)$ m/s

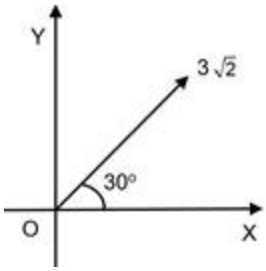
$\left|\overrightarrow{v_u}\right| = \sqrt{36+16}$

= $\sqrt{52}$ m/s

(32) Answer : (2)

Solution:





Y – component is $3\sqrt{2}\ \sin 30^o$

= $3\sqrt{2}\times\frac{1}{2} = \frac{3}{\sqrt{2}}$

(33) Answer : (1)

Solution:

Walking needs static friction to move.

Without friction, foot cannot grip the surface and his foot cannot push against the ground. His foot slides backward instead.

(34) Answer : (3)

Solution:

$\tan\theta = \frac{v^2}{Rg}$

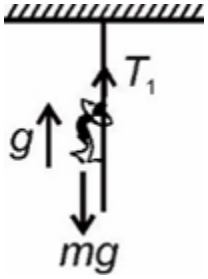
$\tan 53^\circ = \frac{100^2}{R\times 10} \Rightarrow \frac{4}{3} = \frac{1000}{R}$

R = 750 m

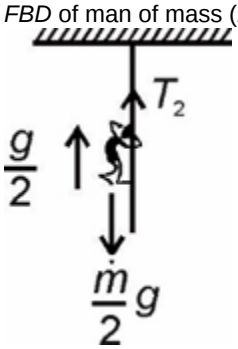
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$\boxed{d = 2R = 1500\ \text{m}}$

(35) Answer : (4)
Solution:
FBD of man of mass (*m*)



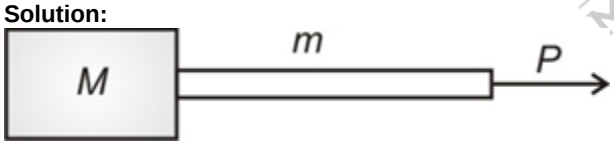
$T_1 = mg + mg$
$T_1 = 2mg$
FBD of man of mass (*m*/2)



$T_2 = \frac{m}{2}g + \frac{m}{2}\cdot\frac{g}{2}$
$T_2 = \frac{mg}{2}\left[\frac{3}{2}\right] = \frac{3mg}{4}$
⇒ $T_1 : T_2 :: 8 : 3$

(36) Answer : (3)
Solution:
Net forces on both the blocks are zero and hence acceleration is zero
$T = 10g = 100$ N

(37) Answer : (3)
Solution:



Acceleration = $\frac{P}{M+m}$
Force exerted by rope = $\frac{MP}{M+m}$

(38) Answer : (4)
Solution:
If $\vec{F}$ is parallel or antiparallel to motion, then it changes magnitude of $\vec{v}$ only but not direction.
If $\vec{F}$ is $\perp$ to motion, it changes direction of $\vec{v}$ and not magnitude of $\vec{v}$
If $\vec{F}$ acts at an angle to motion, it changes both magnitude and direction of $\vec{v}$

(39) Answer : (1)
Solution:
Velocity $\left[\frac{\Delta x}{\Delta t} = \frac{2-0}{2-0} = 1\ \text{m/s}\right]$ at t = 2 s changes from
1 m/s to –1 m/s
Magnitude of impulse $j = \Delta p = 2mv = 2 \times 0.5 \times 1 = 1\ \text{N s}$

(40) Answer : (3)

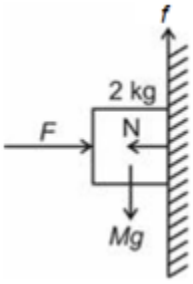
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Solution:
Impulse may or may not change the position of body during the impulsive force.

(41) **Answer :** (2)
Solution:
Limiting friction is always greater than kinetic friction, static friction may be less than kinetic friction.

(42) **Answer :** (3)
Solution:
$\frac{T}{\sin 90^\circ} = \frac{20}{\sin 135^\circ}$
$T = 20\sqrt{2}\text{N}$

(43) **Answer :** (3)
Solution:



In horizontal equilibrium
$N - F = 0$
$N = F$
In vertical equilibrium
$f = \mu F = Mg$
$F = \frac{2\times 10}{0.4} = 50\text{ N}$

(44) **Answer :** (4)
Solution:
$F = m \times a$
$\sqrt{6^2 + 8^2 + 10^2} = m \times 1$
$10\sqrt{2} = m$

(45) **Answer :** (3)
Solution:
2T = (60 + 40) g
2T = 100g ⇒ T = 50 g



$T + N$ = 60 g
N = 60g – 50g = 10 g

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CHEMISTRY

(46) **Answer :** (2)
Solution:
Element B is fluorine.

(47) **Answer :** (2)
Solution:

The IUPAC name of element with atomic number 104 is Unnilquadium

(48) **Answer :** (3)
Solution:

Elements	$\Delta_{eg}H$(kJ/mol)
H	–73
S	–200
Ne	+116
O	–141

(49) **Answer :** (1)
Solution:
Isoelectronic species have same number of electrons.
CO, N_2, CN^- => All these species have 14 electrons.

(50) **Answer :** (2)
Solution:
Atomic size: P>Cl>N>O

(51) **Answer :** (4)
Solution:
Lithium is diagonally related with magnesium

(52) **Answer :** (3)
Solution:
According to Dobereiner's traid, the middle element of each of the triads had an atomic weights about half way between the atomic weights of the other two elements. Dobereiner's traid has similar physical and chemical properties.

(53) **Answer :** (1)
Solution:
$Na \rightarrow Na^+ + e^-$ E = 5.1 eV
$\therefore Na^+ + e^- \rightarrow Na$ E = –5.1 eV

(54) **Answer :** (1)
Hint:
Dubnium has atomic number 105 and has IUPAC symbol Db.
Solution:

	IUPAC official name and IUPAC symbol	Atomic number	Name according to IUPAC nomenclature
a.	Dubnium (Db)	105	Unnilpentium
b.	Seaborgium (Sg)	106	Unnilhexium
c.	Bohrium (Bh)	107	Unnilseptium
d.	Hassium (Hs)	108	Unniloctium

(55) **Answer :** (1)
Solution:
For isoelectronic species, the ionic radii decreases as the atomic number increases.

(56) **Answer :** (2)
Solution:
As is a *p*-block element. While Zn, Cu and Fe are d-block elements, s and p block elements are representative elements.

(57) **Answer :** (4)
Solution:
Atomic number 60 belongs to *f*-block. Elements from 58 – 71 (4*f* series), all lanthanoids, are part of 3rd group.
$(n-2)f^{1-14}\ (n-1)\ d^{0-1}ns^2$ $\Rightarrow$Electronic configuration of *f*-block element.

(58) **Answer :** (2)
Hint:
$E = \frac{hc}{\lambda}$
Solution:
• Power of bulb = 100 watt = 100 Js^{-1}

• Energy of one photon, E = $\frac{hc}{\lambda}$

= $\frac{6.6\times10^{-34}\times3\times10^{8}}{400\times10^{-9}}$

= 4.9 × 10^{-19} J

• Number of photons emitted = $\frac{100\ J}{4.9\times10^{-19}}$

= 2 × 10^{20}

(59) **Answer :** (1)
Solution:

K.E of electron $= 13.6 \times \frac{Z^2}{n^2} \Rightarrow K.\ E \propto n^{-2}$

Angular momentum $= mvr = \frac{nh}{2\pi}$

i.e. Angular momentum ∝ n

Radius of nth orbit $= 0.529 \times \frac{n^2}{Z}$

i.e. $r_n \propto n^2$

Velocity of electron in n^{th} orbit $= 2.188 \times \frac{Z}{n} \times 10^6 m/s$

i.e. $v_n \propto n^{-1}$

(60) **Answer :** (3)
Solution:
• For 4f, n = 4
• For f subshell, l = 3
• Possible values of m_l = –3, – 2, –1, 0, +1, +2, +3
• s = + 1/2 or –1/2

(61) **Answer :** (2)
Solution:

$\lambda = \frac{h}{mv} = \frac{6.6\times10^{-34}}{3.3\times10^{-6}\times200} = 10^{-30}\ m$

(62) **Answer :** (4)
Solution:
p orbital can accommodate maximum two electrons with opposite spin.

(63) **Answer :** (4)
Hint:

$h\nu = h\nu_0 + \frac{1}{2}m_e v^2$

Solution:
If the frequency of light used is greater than the threshold frequency only then electrons are ejected from metal.

(64) **Answer :** (2)

Solution: $E_n^o \frac{Z^2}{n^2}$

(65) **Answer :** (2)
Solution:
Radial nodes = (n – l – 1)
for 3*s*, (3 – 0 – 1) = 2 ; For 3*p*, (3 – 1 – 1) = 1

(66) **Answer :** (1)
Solution:

$\Delta P = 10^{-5}$ kg ms^{-1}

$\Delta x \times \Delta P = \frac{h}{4\pi}$

$\Delta x = \frac{6.6\times10^{-34}\ J}{10^{-5}\times4\times3.14} = 5.2\times10^{-30}\ m$

(67) **Answer :** (3)
Solution:

$r_n = n^2 r_0 \Rightarrow r_3 = (3)^2 x = 9x.$

(68) **Answer :** (1)
Solution:

$\bar{\nu} = \frac{1}{\lambda} = \frac{1}{400\times10^{-7}} = 2.5\times10^{4}\ cm^{-1}$

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(69) Answer : (1)
Solution:
First time of Paschen series $n_1 = 3$, $n_2 = 4$

$\frac{1}{\lambda_1} = \mathrm{R}\left[\frac{1}{9} - \frac{1}{16}\right]$

$\frac{1}{\lambda_1} = \frac{7\mathrm{R}}{144} \Rightarrow \lambda_1 = \frac{144}{7\mathrm{R}}$

Second line of Paschen series $n_1 = 3$, $n_2 = 5$

$\frac{1}{\lambda_2} = \mathrm{R}\left[\frac{1}{9} - \frac{1}{25}\right]$

$\frac{1}{\lambda_2} = \frac{16\mathrm{R}}{225} \Rightarrow \lambda_2 = \frac{225}{16\mathrm{R}}$

$\frac{\lambda_1}{\lambda_2} = \frac{144}{7\mathrm{R}} \times \frac{16\mathrm{R}}{225} = \frac{2304}{1575} = \frac{256}{175}$

(70) Answer : (2)
Solution:
The electron configuration of Ca is: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$.$m_l = 0$ value corresponds to the s-orbitals and each p-subshell has one orbital with $m_l = 0$ that can hold 2 electrons.

(71) Answer : (3)
Solution:
$\frac{1}{\lambda} = R_H\left(\frac{1}{1^2} - \frac{1}{2^2}\right)$

or, $\frac{1}{\lambda} = \mathrm{R_H}\frac{3}{4}$

or, $\lambda = \frac{4}{3\mathrm{R_H}}$

(72) Answer : (3)
Solution:

	List-I (Quantum Numbers)		List-II (Optical)
	n	l	
A.	2	1	2p
B.	4	0	4s
C.	5	3	5f
D.	3	2	3d

l represents the subshell, for which the values are as following :

l	Subshell
0	s
1	p
2	d
3	f

(73) Answer : (4)
Solution:
• For Lyman series, $n_1 = 1$ and $n_2 = 2, 3$ …. Since electron jumps to first excited state (n = 2) hence no line will be observed in Lyman series.
• For Paschen series, $n_1 = 3$ and $n_2 = 4, 5$ …. Hence number of lines obtained will be 3.
• Total number of lines obtained will be 10.

(74) Answer : (4)
Solution:
The shapes of $\mathrm{d}_{xy}, \mathrm{d}_{yz}, \mathrm{d}_{xz}$ and $\mathrm{d}_{x^2-y^2}$ are similar to each other, whereas d_{z^2} is different from others.
They have four lobes except d_{z^2}

(75) Answer : (3)
Hint:
Number of atoms = Moles × N_A × Atomicity
Solution:
a. $\frac{1.8}{18} \times \mathrm{N_A} \times 3 = 0.3\ \mathrm{N_A}$

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b. $\frac{3.1}{31\times4}\times N_A\times4=0.1\ N_A$

c. $\frac{1.6}{32\times8}\times N_A\times8=0.05\ N_A$

d. $\frac{9.2}{46}\times N_A\times3=0.6\ N_A$

(76) Answer : (4)

Hint:

$CaCO_3 + H_2SO_4 \rightleftharpoons CaSO_4 + H_2O + CO_2$

Solution:

Number of moles of H_2SO_4 = Number of moles of $CaCO_3$

$= \frac{10}{100} = \frac{1}{10}$

So, mass of $H_2SO_4 = \frac{1}{10} \times 98\ g = \frac{49}{5} g$

Therefore, mass of solutions

$= \frac{49}{5} \times \frac{100}{60} g$

= 16.33 g

(77) Answer : (3)

Hint:

O_2 is limiting reagent

Solution:

$$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$$

Mole	2		2	
	$\frac{2}{4}$		$\frac{2}{5}$	

So, O_2 is limiting reagent

5 mole of O_2(g) produce → 4 mole of NO(g)

2 mole of O_2(g) produce → $\frac{4}{5} \times 2 = \frac{8}{5}$ = 1.6 mol of NO(g)

5 mole of O_2(g) produce → 6 mole of H_2O(g)

2 mole of O_2(g) produce → $\frac{6}{5} \times 2$ = 2.4 mole of H_2O(g)

(78) Answer : (1)

Hint:

Equivalents of $H_2C_2O_4 \cdot 2H_2O$ = Equivalents of NaOH

Solution:

Molarity of $H_2C_2O_4 \cdot 2H_2O = \frac{12.6\times1000}{126\times1000}$ = 0.1 M

Normality of $H_2C_2O_4 \cdot 2H_2O$ = 0.2

0.2 × 10 = 0.1 × V

V = 20 mL

$$\frac{\frac{(W)_{N_2}}{(MM)_{N_2}}}{\frac{(W)_{O_2}}{(MM)_{O_2}}} = \frac{(n)_{N_2}}{(n)_{O_2}}$$

$$\frac{(n)_{N_2}}{(n)_{O_2}} = \frac{\frac{1}{28}}{\frac{2}{32}} \Rightarrow \frac{16}{28} \Rightarrow \frac{4}{7}$$

(79) Answer : (4)

Hint:

Percentage of oxygen = $\frac{\text{Mass of oxygen}}{\text{Molar mass of compound}} \times 100$

Solution:

a. Percentage of oxygen in H_2SO_4 = $\frac{64}{98} \times 100 = 65.3$

b. Percentage of oxygen in ammonium sulphate

= $\frac{64}{132} \times 100 = 48.48$

c. Percentage of oxygen in SO_2 = $\frac{32}{64} \times 100 = 50\%$

d. Percentage of oxygen in CaO = $\frac{16}{56} \times 100 = 28.57\%$

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(80) **Answer :** (4)
Hint:
Multiple for giga is 10^9
Solution:

Prefix	Multiple
Mega	10^6
Giga	10^9
Exa	10^{18}
Tera	10^{12}

(81) **Answer :** (2)
Solution:
Ratio of moles and balancing coefficient is minimum for B
∴ Limiting reagent is B

(82) **Answer :** (2)
Hint:
$\text{Molality} = \frac{\text{Mole of glucose}}{\text{Mass of water(kg)}}$
Solution:
Mass of glucose = 36 g
Mass of water = (100 – 36) = 64 g
$\text{Molality} = \frac{36/180}{\frac{64}{1000}}$
$= \frac{36}{180} \times \frac{1000}{64}$
= 3.125 m

(83) **Answer :** (3)
Solution:
$3BaCl_2 + 2Na_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6NaCl$
0.3 mole of $BaCl_2$ combines with 0.2 mole of Na_3PO_4. Therefore, $BaCl_2$ is limiting reactant.
0.3 mole of $BaCl_2$ will produce 0.1 mole of $Ba_3(PO_4)_2$ and 0.6 mole of NaCl.

(84) **Answer :** (2)
Hint:
$\underset{100\text{ g}}{CaCO_3} \rightarrow \underset{56\text{ g}}{CaO} + \underset{44\text{ g}}{CO_2}$
Solution:
56 g of CaO is obtained from 100 g $CaCO_3$
22.4 g of CaO is obtained from
$\frac{100}{56} \times 22.4$ = 40 g $CaCO_3$
∴ Percentage purity of sample = $\frac{40}{50} \times 100 = 80\%$

(85) **Answer :** (3)
Hint:
$Al_2(SO_4)_3 \rightarrow 2Al^{3+} + 3SO_4^{2-}$
Solution:
Molarity of $Al_2(SO_4)_3 = \frac{34.2/342}{500/1000}$ = 0.2 M
Molarity of Al^{3+} ions = 0.2 × 2 = 0.4 M
and molarity of SO_4^{2-} ions = 0.2 × 3 = 0.6 M

(86) **Answer :** (3)
Hint:
Mg is limiting reagent.
Solution:
$2Mg + O_2 \rightarrow 2MgO$
Mole of Mg = $\frac{4}{24} = \frac{1}{6}$

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Mole of O_2 = $\frac{8}{32} = \frac{1}{4}$

Mg is limiting reagent

$\therefore$ Mole of MgO formed = $\frac{1}{6}$

(87) Answer : (3)

Hint:

$\text{Molality} = \frac{\text{Number of moles of solute}}{\text{Mass of solvent in kg}}$

Solution:

• 0.25 molal aqueous solution of urea means that 0.25 moles of urea is dissolved in 1000 g of H_2O

• So mass of urea = 0.25 × 60 = 15 g

• So mass of solution = 1000 +15 = 1015 g = 1.015 kg

$\because$ 1.015 kg of urea solution contains 15 g of urea

$\therefore$ 2.5 kg of urea solution contains $\frac{15}{1.015} \times 2.5$ = 37 g

(88) Answer : (2)

Solution:

Empirical formula is the simplest whole number ratio of elements present in a compound.

Element	%	Mole	Ratio
C	40	$\frac{40}{12} = 3.33$	$\frac{3.33}{3.33} = 1$
H	6.67	$\frac{6.67}{1} = 6.67$	$\frac{6.67}{3.33} = 2$
O	53.3	$\frac{53.3}{16} = 3.33$	$\frac{3.33}{3.33} = 1$

Empirical formula is CH_2O

(89) Answer : (3)

Hint:

Mole fraction of the solute

$\left(x_{\text{aoluts}}\right) = \frac{n_{\text{ealum}}}{n_{\text{eoluts}} + n_{\text{eolvent}}}$

Solution:

0.5 molal solution means 0.5 mole of solute in 1 kg water.

1 kg water $= \frac{1000}{18}$ = 55.55 moles

$\therefore x_{\text{solute}} = \frac{0.5}{0.5+55.55} = \frac{1}{112}$

(90) Answer : (2)

Hint:

% by mass of H_2O

= $\frac{\text{Mass of } H_2O}{\text{Mass of compound}} \times 100$

Solution:

$63 = \frac{x \times 18}{18x + 106} \times 100$

$\therefore \quad \boxed{x = 10}$

BOTANY

(91) Answer : (4)

Solution:

Packaging of material to be transported is done by Golgi complex. The materials synthesized in the ER passes outside the cell membrane through Golgi complex.

(92) Answer : (2)

Solution:

Nerve cells are the longest branched cells of human body.

(93) Answer : (2)

Solution:

The central part of the proximal region of the centriole is known as hub.
The non-staining secondary constrictions at a constant location in some chromosomes form a small fragment known as satellite.

(94) Answer : (3)
Solution:
Outer membrane of chloroplast is relatively more permeable and has porins.
Grana is the active site for light reaction.
Stroma is the site for dark reaction or fixation of CO_2 into sugar.

(95) Answer : (2)
Solution:
In mitochondria, enzymes for electron transport are present in the inner membrane.

(96) Answer : (2)
Hint:
In sub-metacentric chromosome, centromere is present slightly away from the centre.
Solution:
In metacentric chromosome, centromere is present at the centre and is also called isobrachial chromosome.
In acrocentric chromosome, centromere is present very close to one end of chromosome. Telocentric chromosome appears to have a single arm.

(97) Answer : (3)
Solution:
Peroxisomes have enzymes for peroxide biosynthesis.

(98) Answer : (1)
Solution:
Site for rRNA synthesis in eukaryote is nucleolus.

(99) Answer : (3)
Solution:
Proteins show only lateral movement within lipid bilayer of plasma membrane.

(100) Answer : (4)
Hint:
A.V. Leeuwenhoek was the first person who observed few living cells, capable of moving, such as bacteria, protozoa, etc.
Solution:
• Robert Brown – Discovered the nucleus of a cell
• Robert Hooke – Discovered the cell from a thin slice of cork.
• Matthias Schleiden – A German botanist who studied a large number of plants and observed that all plants are composed of different kinds of cells.

(101) Answer : (4)
Solution:
Centriole gives cartwheel appearance.

(102) Answer : (4)
Hint:
Microbodies are single membrane bound vesicles containing various enzymes.
Solution:
Microbodies are not present in nucleoplasm. These are found in the cytoplasm.

(103) Answer : (4)
Solution:
Vacuoles in eukaryotes are bound by a single membrane.

(104) Answer : (2)
Solution:
Translation is the process in which ribosomes of polysome translate the mRNA into proteins. This process occurs in mitochondria, chloroplast and cytosol.

(105) Answer : (3)
Hint:
These structures are tubular and help bacteria to attach with rocks also.
Solution:
Fimbriae are bristle like structures which help bacteria to attach with rocks or host cell.

(106) Answer : (3)
Solution:

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Chromoplasts provide various colours - yellow, orange or red, to the plant parts.

(107) Answer : (3)
Solution:
The beginning of diplotene is recognised by the dissolution of synaptonemal complex and the recombined homologous chromosome separate except at the site of crossovers. These X shaped structures are called chiasmata.

(108) Answer : (2)
Solution:
G_1 phase - Interval between most active phase of cell cycle and initiation of DNA replication.
Prophase - Chromosomal material becomes untangled during the process of chromatin condensation.
Telophase - Chromosome cluster at opposite spindle poles and their identity is lost as discrete element.

(109) Answer : (3)
Solution:
Two sequential cycles of nuclear and cytoplasm divisions but single cycle of DNA replication occur in meiosis.

(110) Answer : (4)
Solution:
Bivalents are clearly visible at pachytene. Chiasmata becomes visible during diplotene. Syncytium is a multinucleated condition in coconut.

(111) Answer : (1)
Solution:
Number of chromosomes remains constant before and after the division in mitosis. Although in S phase DNA amount gets doubled but number of chromosomes remains the same.

(112) Answer : (4)
Solution:
Due to meiosis, variations occur from one generation to the next.

(113) Answer : (3)
Solution:
Replication of DNA occurs only once, *i.e.*, in S-phase of cell cycle.

(114) Answer : (3)
Solution:
Cytokinesis is the last stage of M phase. In telophase, the chromosomes reach their respective poles, decondense and lost their individuality.

(115) Answer : (1)
Solution:
Disc-shaped proteinaceous structure associated with centromere is called kinetochore. Kinetochores allow attachment of spindle fibres on the chromosome during cell-division. The purpose of kinetochore is to pull the chromatids apart.

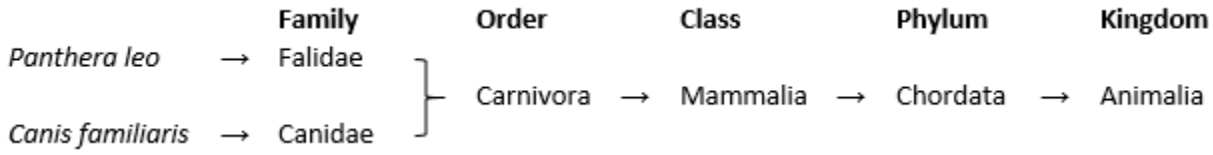
(116) Answer : (4)
Solution:
Sister chromatids separates from each other during anaphase-II of meiosis.

(117) Answer : (2)
Hint:
It is preparatory phase of cell cycle.
Solution:
Interphase is preparatory phase of cell cycle known as resting phase because cell duplicates its organelles and replicates its DNA during this phase, but actual division takes place during M-phase.

(118) Answer : (4)
Solution:
1 → 2 → 4 → 8 → 16 → 32 → 64 → 128
I II III IV V VI VII
7 generations

(119) Answer : (2)
Solution:
Classification of organisms into various categories help in identifying similarities and dissimilarities among different kinds of organism.

(120) Answer : (3)
Solution:



(121) Answer : (4)
Hint:
Triticum is the genus and Angiospermae is the division.
Solution:
The correct ascending order of taxonomic categories is as follows:
Triticum → Poaceae → Poales → Monocotyledonae → Angiospermae

(122) Answer : (4)
Solution:
A division is group of related classes.

(123) Answer : (4)
Hint:
Taxonomy is the branch of science which deals with study of principles and procedures of classification.
Solution:
Modern taxonomy is based upon study of both external and internal features of organism along with cell structure, developmental process and ecological informations of organisms. But phylogeny is included in systematics.

(124) Answer : (3)
Solution:
Poales, Polymoniales – Order
Anacardiaceae – Family
Monocotyledonae – Class
Thus highest rank is Monocotyledonae.

(125) Answer : (4)
Solution:
Binomial epithet includes generic name, specific epithet and author citation.

(126) Answer : (1)
Solution:
Coccus is spherical bacterium.

(127) Answer : (2)
Solution:
According to R.H. Whittaker, all prokaryotes are in kingdom Monera.

(128) Answer : (4)
Hint:
Domains Bacteria and Archaea includes Prokaryotes.
Solution:
Domain Eukarya includes Fungi, Plantae, Protista and Animalia.

(129) Answer : (1)
Hint:
Both archaebacteria and eubacteria are prokaryotes.
Solution:
Archaebacteria differ from other bacteria in having different cell wall structure which is responsible for their survival in extreme conditions.

(130) Answer : (3)
Solution:
Blue green algae are prokaryotes.

(131) Answer : (4)
Hint:
Neutral solutes move across the membrane, by the process of simple diffusion along the concentration gradient.
Solution:
Neutral solute can move across the membrane without any requirement of energy and this is called the passive transport.

(132) Answer : (3)
Solution:

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In some prokaryotes like cyanobacteria, there are other membranous extensions into the cytoplasm called chromatophores which contain pigments.

(133) Answer : (3)

Hint:

Cell division will occur for eight times.

Solution:

Number of generations $= \frac{160}{20} = 8$

Number of cells = $2^8 \times 10^3$ per ml.

= 256×10^3 cells/ml

(134) Answer : (3)

Solution:

Duplication of Golgi apparatus occurs during the G_2-phase of cell cycle.

(135) Answer : (3)

Hint:

The actual period of cell division is M phase.

Solution:

Mitosis is the equational division in which the chromosome number of the parent is conserved in the daughter cell. Meiosis I is a reductional division. Interphase is further subdivided into G_1, S and G_2. S phase is also characterised by chromosome duplication.

ZOOLOGY

(136) Answer : (2)

Solution:

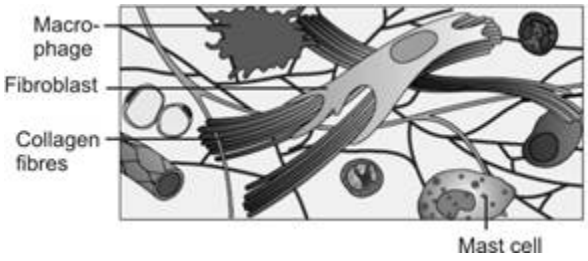
PCT of nephron	–	Brush bordered cuboidal epithelium
Bronchioles	–	Ciliated columnar epithelium
Stomach and intestine	–	Columnar epithelium
Wall of blood vessels	–	Simple squamous epithelium

(137) Answer : (1)

Hint:

Areolar tissue

Solution:



(138) Answer : (3)

Solution:

Tight junctions help to stop substances from leaking across a tissue. Adhering junctions perform cementing to keep neighbouring cells together.

(139) Answer : (3)

Solution:

Smooth muscle fibres are elongated and spindle-shaped *i.e.*, pointed or tapered at the ends (fusiform) and are uninucleated.

(140) Answer : (4)

Solution:

In heart, kidney and capillaries cartilage is absent.

(141) Answer : (1)

Solution:

Unicellular glands contain isolated glandular cells, *e.g.*, goblet cells in alimentary canal which secrete mucus.

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(142) Answer : (3)

Solution:

Tendon connects skeletal muscle to a bone and is dense regular connective tissue.

(143) Answer : (4)

Hint:

Compound epithelium

Solution:

Compound epithelium is made of more than one layer (multi-layered) of cells and thus, has a limited role in secretion and absorption. Their main function is to provide protection against chemical and mechanical stresses. They cover the dry surface of the skin, the moist surface of buccal cavity, pharynx, inner lining of ducts of salivary glands and of pancreatic ducts.

(144) Answer : (4)

Solution:

In humans, the simple squamous epithelium is usually found in the walls of blood vessels, air sacs of lungs and Bowman's capsule.

(145) Answer : (2)

Hint:

Porifera show cell aggregate body plan and tissues are lacking.

Solution:

Members of the phylum Cnidaria/Coelenterata exhibit tissue level of organisation for the first time in kingdom Animalia. The members of the phylum Platyhelminthes and Aschelminthes exhibit organ and organ system level of organisation respectively.

(146) Answer : (1)

Solution:

Neurotransmitters are either excitatory or inhibitory.

Arrival of the signal at neuron endings can result in stimulation or inhibition of adjacent neurons depending on the type of neurotransmitter released.

(147) Answer : (2)

Solution:

Given tissue is bone as it possesses haversian canals and concentric lamellae.

(148) Answer : (4)

Solution:

Neurons are the structural unit of nervous tissue.

(149) Answer : (2)

Solution:

Microvilli are found in the lining of intestine.

(150) Answer : (1)

Solution:

A part of pharynx is the common passage for food and air.

(151) Answer : (2)

Solution:

Ribozymes are RNA molecules which act as enzymes.

(152) Answer : (4)

Solution:

Collagen is the most abundant protein in the animal world and Ribulose Bisphosphate Carboxylase-Oxygenase (RuBisCO) is the most abundant protein in the whole of the biosphere.

(153) Answer : (1)

Hint:

Water is the most abundant chemical in living tissue.

Solution:

Component	% of the total cellular mass
Water	70-90
Proteins	10-15
Carbohydrates	3
Lipids	2
Nucleic acids	5-7
Ions	1

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(154) Answer : (1)

Solution:

Based on Chargaff's rule

A + G = T + C

[A] = [T] & [G] = [C]

C = 20%, A = 30% and T = 30%

(155) Answer : (2)

Solution:

The effect of a competitive inhibitor is reversed by increasing substrate concentration. A competitive inhibitor increases the apparent K_m for a given substrate.

(156) Answer : (4)

Solution:

The acid-insoluble fraction contains compounds with molecular weight higher than ten thousand daltons, except lipids.

(157) Answer : (4)

Solution:

Catalytic activity is lost when the co-factor is removed from the enzyme which testifies that they play a crucial role in the catalytic activity of the enzyme.

At low temperature, enzymes become inactive.

(158) Answer : (3)

Solution:

Antibody fights against infectious agents. It is used by the immune system to identify and neutralize foreign objects/antigens.

(159) Answer : (1)

Solution:

All amino acids can exist in zwitterionic form at some stage in a living cell. A dipeptide has one peptide bond.

(160) Answer : (2)

Hint:

Equal to the number of OH groups in a molecule of glycerol.

Solution:

Heterocyclic ring means presence of any other atom apart from carbon in that ring.

From the mentioned compounds, ribose, adenylic acid and uracil possess heterocyclic ring.

Serine is an amino acid and does not possess cyclic ring.

Glycerol is trihydroxypropane.

Cholesterol has homocyclic ring.

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(161) Answer : (3)

Hint:

Homopolymer of glucose

Solution:

Cellulose is a homopolysaccharide consisting of only one type of monosaccharide units *i.e.,* glucose. Unlike starch, cellulose does not contain complex helices and hence, cannot hold I_2 molecules. Since it cannot hold I_2, it remains colourless during the I_2 test.

(162) Answer : (4)

Solution:

GLUT-4 is a protein and it contains peptide bonds. Insulin is the protein hormone.

A glycosidic bond exists in the RNA molecule between sugar and nitrogen base.

(163) Answer : (3)

Hint:

Phenylalanine is an aromatic amino acid.

Solution:

The chemical and physical properties of amino acids are essentially of the amino, carboxyl and the R functional groups. Based on number of amino and carboxyl groups, there are acidic (glutamic acid), basic (lysine) and neutral (valine) amino acids. Similarly, there are aromatic amino acids (*e.g.*, tyrosine, phenylalanine, tryptophan).

(164) Answer : (4)

Solution:

DNA is the genetic material in humans. DNA has deoxyribose sugar while RNA has ribose sugar. DNA has cytosine and thymine while RNA has cytosine and uracil.

(165) Answer : (2)

Hint:

Connects adjacent nucleotides

Solution:

In nucleic acids, one ester bond is present on each side of phosphate group therefore, it is called phosphodiester bond. Peptide bond links amino acids in a polypeptide. Glycosidic bond links individual monosaccharides in a polysaccharide. Hydrogen bonds stabilize the tertiary structure of proteins.

(166) Answer : (1)

Solution:

Option (1) is the correct answer because statements (c), (d) and (e) are correct. Oils have lower melting point and hence remain as oil in winters. Lipids are generally insoluble in water but soluble in some organic solvents like chloroform or benzene.

Option (2), (3) and (4) are incorrect because statements (a) and (b) are incorrect. Lecithin is a type of phospholipid found in plasma membrane. Saturated fatty acids are without double bond (C = C).

(167) Answer : (2)

Solution:

Haem is a prosthetic group for enzymes like catalase and peroxidase. Vitamins are essential components of many co-enzymes like NAD and NADP. Zinc is a co-factor for the proteolytic enzyme, carboxypeptidase.

(168) Answer : (2)

Solution:

Functional residual capacity is defined as the volume of air that will remain in the lungs after a normal expiration. This includes expiratory reserve volume and residual volume.

FRC = ERV + RV = 2300 mL

(169) Answer : (1)

Solution:

During inspiration, the volume of the thoracic cavity increases. This increase occurs in the antero-posterior axis due to the contraction of the diaphragm. During expiration, decrease in the volume of the thoracic cavity in the dorso-ventral axis and antero-posterior axis is seen.

(170) Answer : (3)

Solution:

At cellular level, O_2 is used to breakdown glucose into CO_2, water and generates ATP which is used to provide energy for various processes.

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(171) Answer : (2)

Solution:

Labeo – Gills
Pheretima – Moist cuticle
Periplaneta – Tracheal tubes
Pteropus – Lungs

(172) Answer : (3)

Solution:

Breathing or pulmonary ventilation is first step of respiration by which atmospheric air is drawn in and CO_2 rich alveolar air is released out.

(173) Answer : (3)

Solution:

Larynx is located anterior to the oesophagus. Its upper part has an opening, called glottis. It is a part of the conducting zone of the respiratory passage.

(174) Answer : (1)

Solution:

The thoracic chamber is formed ventrally by the sternum, laterally by the ribs and on the lower side (inferiorly) by the dome-shaped diaphragm.

(175) Answer : (3)

Solution:

Cutaneous respiration occurs through skin. Pulmonary respiration occurs through lungs.

(176) Answer : (4)

Solution:

Breathing is a mechanical process while respiration is a chemical/metabolic process.

(177) Answer : (1)

Solution:

Alveolar ducts and respiratory bronchioles are the parts of the respiratory zone of the respiratory system.
Primary bronchus undergoes repeated divisions to form secondary, tertiary bronchi and bronchioles.

(178) Answer : (4)
Solution:
Pleura splits into two layers outer parietal pleura invested on chest wall and inner visceral pleura invested on lung surface.
Pleural fluid is present between two pleura to reduce friction during movement of lungs.

(179) Answer : (2)
Solution:

Trachea divides at the level of 5th thoracic vertebra into a right and left primary bronchi.

(180) Answer : (4)
Solution:
Spirometer can evaluate respiratory volumes to indicate functioning of the lung.

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